

# **SCHEME Interpreter Chip**

**by**

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```

(define (read-eval-print-loop)
  (terpri)
  (princ '|==> |)
  (compile-to-++ (read))
  (aset '** (++))
  (print **))
  (read-eval-print-loop))

(#CLOSURE
 (#PROCEDURE
  (#SEQUENCE
    (#APPLY-NO-ARGS (#GLOBAL TERPRI) NIL)
    (#SEQUENCE
      (#APPLY-1-ARG (#SELF-EVALUATING-IMMEDIATE |==> |)
        (#GLOBAL PRINC))
      (#SEQUENCE
        (#APPLY-1-ARG (#APPLY-NO-ARGS (#GLOBAL READ) NIL)
          (#GLOBAL COMP-TO-++))
        (#SEQUENCE
          (#SEQUENCE (#APPLY-NO-ARGS (#GLOBAL ++)) NIL)
            (#SET-GLOBAL **))
          (#SEQUENCE
            (#APPLY-1-ARG (#GLOBAL **) (#GLOBAL PRINT))
            (#APPLY-NO-ARGS (#GLOBAL READ-EVAL-PRINT-LOOP)
              NIL))))))
    (#SYMBOL READ-EVAL-PRINT-LOOP))
  NIL)

```

```

(define (app x y)
  (cond ((eq x 'nil) y)
        (t (cons (car x)
                   (app (cdr x) y)))))

(#CLOSURE
 (#PROCEDURE
  (#SEQUENCE
    (#FIRST-ARGUMENT (#LOCAL (0 . 0)))
    (#PRIMITIVE-APPLY-2 (#SELF-EVALUATING-IMMEDIATE NIL)
                        (#PRIMITIVE-EQ NIL)))
  (#CONDITIONAL (#LOCAL (0 . 1))
    (#FIRST-ARGUMENT
     (#PRIMITIVE-APPLY-1 (#LOCAL (0 . 0))
                          (#PRIMITIVE-CAR NIL))
     (#PRIMITIVE-APPLY-2
      (#FIRST-ARGUMENT
       (#PRIMITIVE-APPLY-1 (#LOCAL (0 . 0))
                            (#PRIMITIVE-CDR NIL))
       (#LAST-ARGUMENT (#LOCAL (0 . 1))
                        (#GLOBAL APP)))
      (#PRIMITIVE-CONS NIL))))))
  NIL)
NIL)

```